

Privacy Preserving Federated Learning for Lung Cancer Classification

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Abstract—Lung cancer is characterized by high mortality and incidence rates, making it one of the most prevalent cancers globally. Early detection significantly improves the chances of survival for individuals affected by this disease. The histopathological diagnosis is a crucial factor in determining the specific type of cancer. In recent years, there has been a significant increase in novel computer-aided diagnostic techniques utilizing deep learning algorithms for the early detection of lung cancer. However, sharing sensitive patient data is significantly restricted by regulations such as HIPAA and GDPR, primarily due to privacy concerns. Given the current constraints, institutions face challenges in effectively exchanging information to enhance the accuracy of lung cancer classification. In order to address the issue of privacy in lung cancer classification, we propose a federated learning approach. This methodology involves employing local models with an Inception-v3 backbone to carry out the classification of histopathological images of lung cancer & updating the global model based on the local weights. These images have been obtained from the LC25000 dataset. The lung cancer images from the LC25000 dataset were analyzed, which consisted of three distinct classes. Each class contained a total of 5000 images. The applied model has achieved a classification accuracy of 99.867% in categorizing lung cancer images into three distinct classes. The performance of the proposed framework has demonstrated superiority over other existing methodologies. Furthermore, this solution effectively addresses the privacy concerns associated with the sharing of medical data among different institutions.

Index Terms—Lung Cancer, Histopathological Images, Image Classification, Decentralized machine learning, Federated learning, Privacy Preservation

I. INTRODUCTION

Cancer is becoming one of the leading causes of death throughout the globe. Based on the most recent data from the Global Cancer Statistics (GLOBOCAN) [1], the global incidence of new cancer cases stands at 19.3 million. In the year 2020, the total number of cancer-related deaths amounted to 9.96 million individuals. In 2020, lung cancer emerged as

the prevailing form of cancer among males, with a staggering 1.43 million newly diagnosed cases. In the female population, there were 0.77 million reported cases, ranking second in prevalence. Breast cancer, with 2.26 million cases diagnosed in 2020 across both genders, held the top position. Lung cancer accounted for the highest number of cancer-related fatalities, resulting in 1.8 million deaths, which represents 18% of all cancer deaths. The projected cancer incidence estimates for the year 2040 are significant, with a global burden of over 28 million cases anticipated. This represents a substantial 47% rise from the figures recorded in 2020 [1].

Furthermore, as indicated by a survey [2], there exists a notable incidence of misdiagnosis in cases of lung cancer. Histological images remain a widely used approach for cancer diagnosis in the medical field. The field of clinical medicine has demonstrated that symptoms of lung cancer typically manifest in the advanced stages of the disease, making it challenging for doctors to diagnose lung cancer early on by relying solely on the visual assessment of CT images. Physicians experience heavy workload as they must review almost 10,000 photos daily. This increases the likelihood of clinician misdiagnosis. The utilization of computer aided diagnosis (CAD) in the analysis of histopathological images has consistently been a significant area of focus within the realm of cancer detection. Numerous studies have been conducted in this field utilizing artificial intelligence (AI) [3].

Regulations like HIPAA [4] and GDPRs [5] make it very difficult for institutions to disclose patient data, even anonymized ones; privacy is a top priority for healthcare facilities. To address these challenges we propose the implementation of a decentralized machine learning approach known as Federated Learning. There are currently multiple deep learning methods available that demonstrate high accuracy in predicting the class of lung cells. However, none

of these entities have implemented a comprehensive approach to address data privacy regulations. The effectiveness of our framework is assessed through a comparative analysis with other methods that have been applied to the same dataset. The proposed methodology has demonstrated enhanced accuracy in addressing lung cancer classification tasks and effectively mitigating the challenges posed by data privacy regulations.

The following is the structure of the parts that follow this introduction. Section II provides a concise summary of previous studies that exhibit commonalities with the current research. Section III presents an overview of the dataset used in this study and the research methodology presented. Section IV of this document is dedicated to offering a comprehensive explanation of the research model that is suggested. In the section V, the results and a comparative analysis are presented. Ultimately, the article concludes in Section VI by providing a concise overview of the results.

II. LITERATURE REVIEW

In recent years, there has been significant progress made by several research groups in the field of lung cancer diagnosis. These advancements involve the utilization of machine learning techniques that are based on the analysis of histopathological images. Hlavcheva et al. [6] utilized different CNN architectures to process lung biopsy images in conjunction with median filtering as a means of pre-processing CT scan data. They achieved an accuracy of 94.6% for distinguishing lung cancer cells. They identified normal and damaged lung images by extracting features using mathematical morphology.

A modified GoogLeNet inception module was integrated with a transfer learning model by Al-Huseiny et al. [7]. Accuracy of 94.38% was attained using the IQ-OTH/NCCD lung cancer dataset. Mangal et al. [8] proposed a neural network-based (CAD) system for distinguishing between the four types of lung cancer using digital pathology images. For the lung and colon, respectively, the diagnosis accuracy was found to be over 97% and 96%. A Convolutional Neural Network (CNN) model for identifying lung cancer was presented by Saif et al. [9], which was trained on data from three different subtypes of lung cancer. The findings showed that their suggested CNN model got an impressively high rate of accuracy, 98.53% overall.

Masud et al. [10] developed a deep learning (DL) supervised learning approach to categorize lung cancer tissues into five distinct groups. In their methodology, they use two different feature extraction approaches. Overall, they were successful to in achieving an accuracy of 96.33 %. In order to categorize CT scans of lung cancer, Talukder et al. [11] proposed a method that places high performance filtering and ensemble feature extraction. Ensemble learning model a good feature extraction approach led them to a remarkable 99.05% accuracy. In order to evaluate the efficacy of transfer learning models and a Hybrid CNN methodology, Sultana et al. [12] conducted a comparison between the two. The outcomes showed that the transfer learning method had a 99.13% success rate.

The analysis of medical images has come a long way, especially with the advent of decentralized machine learning. Decentralized healthcare technology was the subject of an in-depth study published by Nicola et al. [13]. The writers of this viewpoint article set out to educate readers on the merits and effects of FL (Federated Learning). Moreover, they presented a unique idea for a decentralized federated learning (FL) architecture that makes use of blockchain technology. Using the LC25000 lung and colon cancer histopathology dataset, this framework was applied to the job of lung cancer classification using the EfficientNetB7 model. According to reference [14], the results showed an outstanding accuracy of 96.52% on the test set. In addition, Arthi et al. [15] suggested a Federated Learning method using Explainable AI (XAI) to classify colorectal cancer, with an accuracy of 99.53%.

III. METHODOLOGIES

A. Data Collection

In this research, we made use of the LC25000 dataset, which was released in 2020 by A. Borkowski et al. [16]. This dataset comprises a total of 25,000 images representing lung and colon tissues. These images have been sorted into five clear-cut categories. Specifically, there are three different categories of lung tissue images, which are Adenocarcinoma, Squamous Cell Carcinoma, and Benign. For our study, we exclusively utilized the lung cancer image directory within the dataset, which comprises three subfolders, each holding 5,000 images.

B. Dataset Splitting

The dataset containing 15,000 lung images is organized into training, testing, and validation sets using an 80:10:10 ratio.

TABLE I
DESCRIPTION OF THE EMPLOYED DATASET.

Image Type	Folder Title	Total Images	Train Set	Test Set	Validation Set
Lung Adenocarcinoma	lung_aca	5000	4000	500	500
Lung Benign	Lung_bnt	5000	4000	500	500
Lung Squamous Cell Carcinoma	lung_scc	5000	4000	500	500

C. Dataset Preprocessing and Augmentation

Initially, 1250 images of lung and colon tissues were included in the LC25000 dataset. A. Borkowski et al. [16] used strategies for flipping and rotating the samples in a wide range of orientations to improve the dataset. Because of this, the size of the dataset grew to a total of 25,000 images, which were then split up into five groups of 5,000. All pictures were reduced to 768 pixels on the widest and highest dimensions before augmentation techniques were applied. The images were carefully verified and made freely available to guarantee privacy and HIPAA compliance. Figure 1 shows several examples from the collection, and Table 1 lists the labels and identifiers for the various picture types in the dataset.

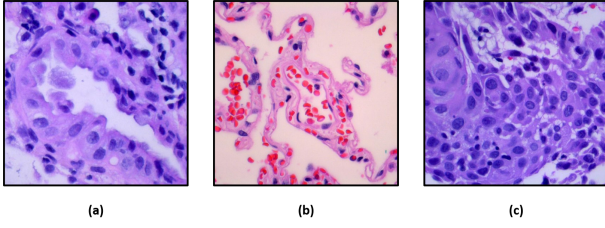


Fig. 1. Sample Images of LC25000 dataset. (a) Lung Adenocarcinoma. (b) Lung Benign. (c) Lung Squamous Cell Carcinoma.

D. Federated Learning

Federated Learning is a machine learning methodology that enables multiple devices, known as clients, to collectively train a shared model without the need to transmit their raw data to a central server. In this scenario, the participants engage in the exchange of model updates or weights while ensuring that their data remains localized. Stochastic Gradient Descent based Federated Learning (FedSGD) is used to perform local model training. The process involves training local models on individual client devices using their own data for a single epoch. The local models are initialized with the weights of the global model, but they may exhibit slight divergence as a result of their unique datasets. Local updates play a crucial role in the implementation of Federated Learning. They serve the purpose of capturing local knowledge while ensuring data privacy by abstaining from sharing raw data with the central server, also known as the global model.

Algorithm 1 Federated Learning algorithm

Require: $n \leq \text{comms_round}$

Ensure: Initialize global model with random weights

- 1: Get the current weights of the global model;
 - 2: **for** $n \leq \text{comms_round}$ **do**
 - 3: Get the current weights of the global model;
 - 4: Collect scaled local weights;
 - 5: Get a list of client_names;
 - 6: **for** each client in client_names: **do**
 - 7: Set the local model's weights to global_weights;
 - 8: Train the local model on client's data;
 - 9: Append the scaled weights;
 - 10: **end for**
 - 11: Calculate the average weights by summation;
 - 12: Update the global model's weights;
 - 13: Evaluate the global model's performance on test data;
 - 14: **end for**
 - 15: End of the algorithm
-

As our clients data were Independent and Identically Distributed Data(IID), Federated Averaging is used when updates from individual clients' local models are aggregated to create a more accurate global model. The Federated Averaging part ensures that the contributions from different clients are combined and balanced, taking into account the size and importance of each client's data. This aggregation step helps create a global

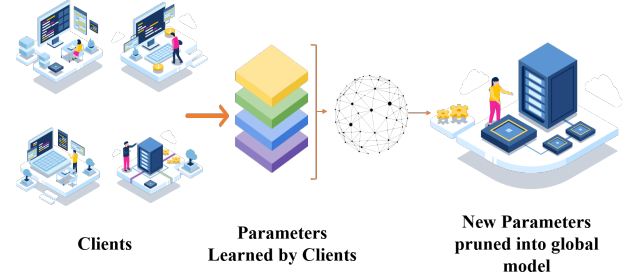


Fig. 2. Locally trained models effectively create a new global model.

model that benefits from the collective knowledge of all clients while preserving data privacy. Fig. 2 demonstrates the process of updating global model through local client parameters.

E. Architecture

Our main objective here is to classify lung cancer images using federated learning. The goal of image classification is to teach a computer program to recognize and assign a label or category to an image based on its visual content. In Order to work with federated learning, we need a local image classification model that can train individual client devices using their own data for a single epoch. For this purpose we have chosen a pretrained convolutional neural network called InceptionV3.

InceptionNet: Google developed a collection of deep convolutional neural network (CNN) designs for computer vision tasks called InceptionNet, or the Inception architecture [17]. The Inception architecture is distinguished by its utilization of Inception modules, which are fundamental components comprising several parallel convolutional layers featuring varying filter sizes. The utilization of these modules enables the network to effectively capture features at different scales and hierarchies present in an image, thereby improving its capability to accurately identify intricate patterns and objects.

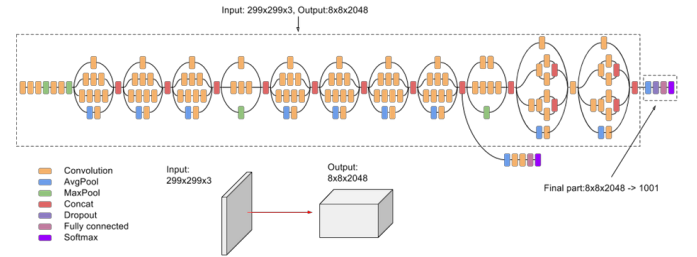


Fig. 3. Architecture of Inception v3 model [17].

The Inception architecture was first introduced alongside the original Inception (GoogLeNet) model, which emerged as the victor in the 2014 ILSVRC Challenge. The number of parameters in this model has been significantly reduced compared to previous architectures such as AlexNet and VGGNet, while still achieving high accuracy. The network can collect multi-scale properties without parameter inflation. The reduction in

computational complexity was achieved by utilizing factorized convolutions, specifically 1x1 and 3x3 convolutions. in Fig 3. we can see the overall architecture of the Inception-v3.

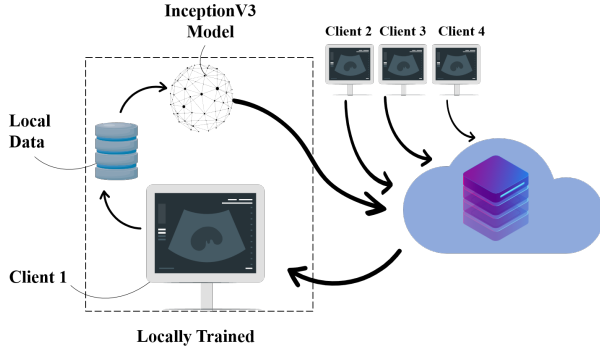


Fig. 4. Workflow for the system to classify lung cancer.

The subsequent iterations of the Inception architecture, namely InceptionV2, InceptionV3, and InceptionV4, brought about enhancements and optimizations. For our task at hand, we have utilized InceptionV3 at the local clients for classification on individual devices. It can be seen from Fig. 4 where our InceptionV3 plays its role. These advancements included the incorporation of batch normalization, different filter factorizations, and the use of deeper networks. The utilization of these models has been extensively prevalent in both research and practical applications, particularly for tasks involving image classification, object detection, and image segmentation.

F. Evaluation Metrics

To determine the effectiveness of the proposed model, a number of assessment criteria were applied. This set of criteria includes the following: specificity, recall, F1-score, accuracy, precision, and confusion matrix. Here, we present the formulas for these metrics.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 - Score = \frac{2 \times (Precision \times Recall)}{Precision + Recall} \quad (4)$$

$$Specificity = \frac{TN}{TN + FP} \quad (5)$$

IV. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

A. Set-Up of Experiments

The experiments were conducted using the Google Colab platform, and the hyperparameters used in the computations are presented in Table II. The proposed Federated learning

Model was run on NVIDIA A100 40 GB GPU and 85 GB RAM provided by Google Colab services. The images for the original dataset were with 100x100 dimensions that were trained in this environment.

TABLE II
HYPERPARAMETER SETTINGS OF PROPOSED MODEL

Hyperparameter	Value
No. of Comm. Round	50
Batch Size	16
No. of Clients	5
Optimizer	Adam
Loss	Categorical Crossentropy

B. Results of Federated Learning with InceptionV3

The section covers an overview of the findings using Federated Learning using Inception-v3 features of the lung cancer dataset's histology images.

The performance of our suggested Federated Learning Model utilizing a Inception-v3 architecture that has already been trained is summarized in Table III. Our experimented dataset contained three classes. Table III illustrates that the model achieved 100% recall, 100% precision, 99.6% accuracy, and 100% f1-score while identifying "lung_aca." The suggested model obtained a 100% score for accuracy, precision, recall, and f1-score for the other two classes, "lung_bnt" and "lung_scc." All things considered, our suggested model has classified lung cancer histology images with an accuracy of 99.867%.

TABLE III
RESULTS OF FEDERATED LEARNING WITH PRETRAINED INCEPTION-V3

Type of Class	Precision	Recall	F1-score	Specificity	Accuracy
lung_aca	1.00	1.00	1.00	1.00	99.6%
lung_bnt	1.00	1.00	1.00	1.00	100%
lung_scc	1.00	1.00	1.00	1.00	100%
Average	1.00	1.00	1.00	1.00	99.867%

The confusion matrix of our proposed federated learning model is shown in Fig. 5 & the global accuracy graph epochs wise is shown in Fig. 6.

V. COMPARATIVE RESULT ANALYSIS

Table IV presents a comparative analysis of the accuracy, precision, and recall of four neural network models and the proposed federated learning model for the classification of lung cancer. After reviewing the methods used to obtain the models' predictions, we compared the results of five different models: Federated Learning with Inception-V3, Inception-V3, VGG-16, ResNeXt50, Xception, and ResNet-50. The findings demonstrate that, with a 99.87% classification accuracy, the federated learning methodology not only allays privacy concerns regarding medical data, but also outperforms previous neural network models.

With an accuracy rate of 99.27%, the Xception model performed the closet to our suggested model in the classification of lung cancer types. Inception-v3, our baseline

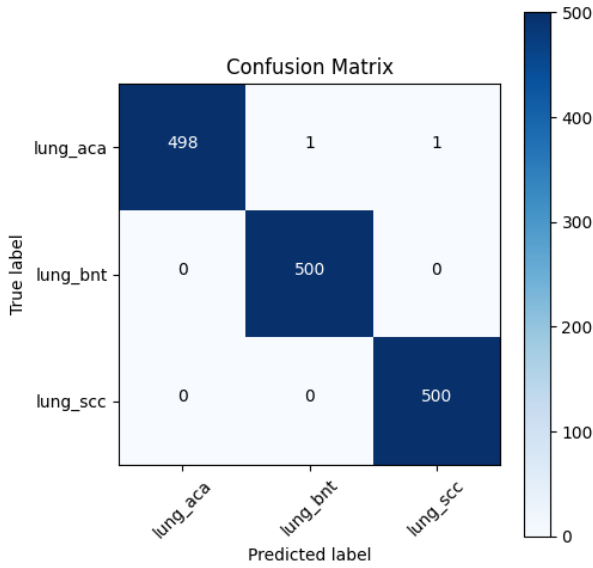


Fig. 5. Confusion Matrix for Lung Cancer Classification.

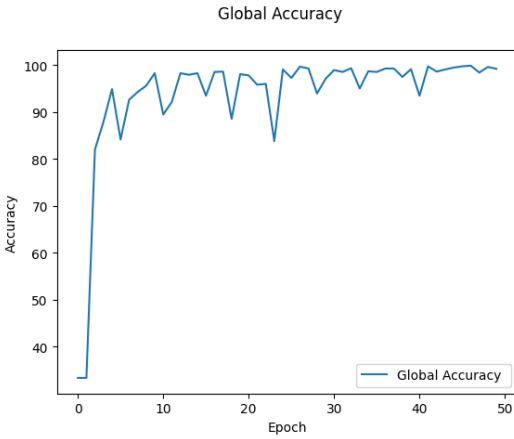


Fig. 6. Global Accuracy Graph for Federated Learning.

architecture, on the other hand, provided us with an accuracy of 99.16%. ResNet-50 outperformed Google's initial model, with a 99.20% accuracy rate. ResNeXt-50, the model that preceded ResNet-50, had good results, achieving a 99.20% classification accuracy. VGG-16, on the other hand, performed the worst out of all the models, with an accuracy of 98.13%. Fig. 7 shows us how the clients individual accuracy varies from the global accuracy.

Based on the findings presented in Table V, numerous studies have been carried out on the LC25000 dataset, demonstrating its relevance to our own research. The analysis presented above clearly demonstrates that the proposed model achieved superior classification results compared to the currently available methods. The proposed method utilizes distributed devices to improve the speed and efficiency of model training. Higher efficiency can be attained by increasing the quantity of data acquired from several clients. Federated

TABLE IV
COMPARISON OF FEDERATED LEARNING WITH OTHER NEURAL NETWORK BASED MODELS

Classification Model	Precision	Recall	Accuracy
Federated Learning with Inception-V3	1.00	1.00	99.867%
Inception-v3	0.9916	0.9916	99.16%
VGG16	0.9833	0.9833	98.33%
ResNet-50	0.9926	0.9920	99.20%
ResNeXt50	0.9920	0.9927	99.20%
Xception	0.9927	0.9927	99.27%

learning eliminates the need to transmit large amounts of data to a central server. In this approach, the server and clients exclusively exchange model updates, also known as gradients. This method proves beneficial in scenarios with limited bandwidth, as it effectively reduces communication costs.

TABLE V
COMPARATIVE ANALYSIS WITH PREVIOUS WORKS ON LC25000 DATASET

Previous Studies	Year	Methodology	Accuracy
Hlavcheva et al. [6]	2021	CNN Model	94.6%
Masud et al. [10]	2021	CNN Model	96.33%
Mangal et al. [8]	2020	CNN Model	97.89%
Karim et al. [18]	2021	CNN Model	98.07%
Shandilya et al. [19]	2022	CNN Model	98.67%
Talukder et al. [11]	2022	Ensemble Learning	99.05%
Sultana et al. [12]	2021	Transfer Learning	99.13%
Chehade et al. [20]	2021	XGBoost	99.53%
Our Method	2023	Federated Learning	99.867%

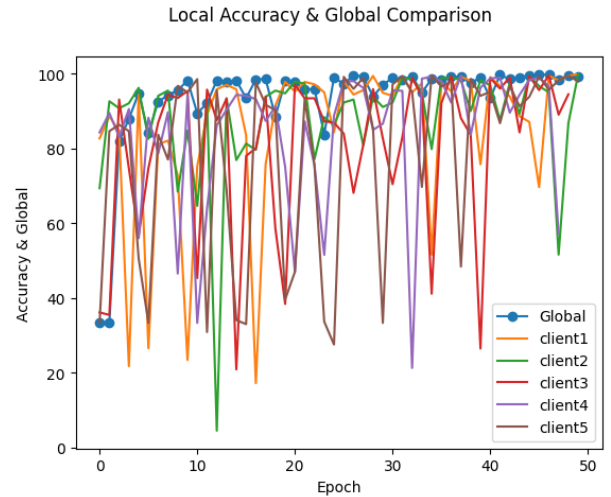


Fig. 7. Global Accuracy v/s Local Accuracy.

VI. CONCLUSION

Lung cancer is a prevalent malignancy that is associated with high mortality rates. Early detection increases the likelihood of survival. However, physicians may encounter difficulties in accurately detecting lung cancer using CT images.

The purpose of this study is to utilize a convolutional neural network (CNN) approach to detect lung cancer at an early stage, while taking into consideration data sharing regulations. This strategy consists of two systems. The initial approach involves utilizing the Inception-v3 model to process local models on individual client devices, leveraging the data specific to each device. The process involves aggregating local models to generate a global model with enhanced accuracy. It is crucial to acknowledge that each individual device maintains ownership of its local data. The global server receives only the parameters and weights. This component is an integral part of the federated learning system, designed specifically to address and mitigate privacy concerns.

The proposed systems demonstrated exceptional performance in the early diagnosis of LC25000 dataset images. The classification accuracy of 99.867% was achieved using the federated learning methodology with the pretrained Inception-v3 features. This method outperforms other cutting-edge techniques. Despite the notable achievements of our proposed model, it is important to acknowledge its inherent limitations, which may impede the training process. The analysis was conducted on IID data, acknowledging the inherent variability in image sizes across different institutions. The management of non-independent and identically distributed (non-IID) data is a significant area of focus. In our upcoming research, we will conduct an evaluation of the existing framework using various datasets. Additionally, we will investigate the model's ability to handle non-independent and identically distributed (non-IID) data.

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